SIMATS ENGINEERING
ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Assessment Tool Weightage: 40% **Total Marks: 30**

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

257

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Code of Conduct (192311326)
I S. Jaiwant (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ____________

Assessment Tasks
Class Test
1. Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.
2. Explain the 5Vs of Big Data with one practical example for any two Vs.
3. Differentiate structured, semi-structured, and unstructured data with examples.
4. Explain the role of distributed file systems in big data processing.
5. List any three major security concerns in big data platforms and briefly explain each.
6. What are data appliances and integration tools? Explain their role in big data workflows.
7. Discuss two major challenges organizations face while implementing big data analytics.
8. Write short notes on any one domain application of big data analytics.
9. How do business drivers influence the adoption of big data solutions?
10. Explain the importance of big data analytics in decision making.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage

Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

CO-4

ASSESSMENT - 2

(1) BIG DATA :-

(*) Big data refers to extremely large and complex datasets that are difficult to store, process, and analyze using traditional system.

(*) Traditional methods cannot efficiently handle the huge volume, velocity and variety of data.

(*) Big data uses distributed storage and parallel processing.

(2) 5Vs of Big Data :-

The 5Vs are :-

(*) Volume - Huge amount of data.

(*) Velocity - High speed data generation.

(*) Variety - Different types of data.

(*) Veracity - Accuracy and reliability.

(*) Value - Useful info obtained from data.

Eg:- Social media generates huge volume, while stock market generates data at high velocity.

3 TYPES OF DATA :-

(*) Structured : Fixed format, such as SQL tables

(*) Semi-structured : Partially organized, such as JSON and XML

(*) Unstructured : NO fixed format, such as images, videos and audio.

4 Distributed file system :-

(*) A Distributed file system stores large datasets across multiple computers (or) nodes.

(*) It divides data into blocks and distributes them across the system.

(*) It provides scalability, fault tolerence and parallel processing.

Example : HDFS.

5 BIG DATA SECURITY :-

The major security concerns are :-

(*) Unauthorized access - Preventing unwanted users from accessing data.

(*) Data privacy - Protecting personal and confidensial information.

(*) Data integrity - Preventing unauthorized modification (or) corruption of data.

6 DATA APPLIANCE AND INTEGRATION TOOLS :-

(*) Data appliances are specialized system used for storing and processing large amounts of data.

(*) Integration tools collect, transform and combine data from different sources.

(*) They support data collection, transformation, storage, and processing.

7 Big DATA IMPLEMENTATION CHALLENGES :-

(*) Security and privacy : Protecting sensitive data from attacks and unauthorized access.

(*) Lack of skilled professional : Big Data requires knowledge of Hadoop, spark, cloud computing and analytics.

⑧ BIG DATA IN HEALTHCARE :-

Big DATA is used to analyze patient records medical images and sensor data. It supports disease prediction, personalized treatment, patient monitoring and hospital management, improving health care.

⑨ BUSINESS DRIVERS :-

It Encourage organization to adopt Big data for Better decision-making, cost reduction, customer understanding, increased revenue and competitive advantages. Business requirements determine how Big data solution are used.

⑩ BIG DATA ANALYTICS IN DECISION MAKING :-

It identifies useful patterns, trends and insights from large data set. & It helps organizations Make faster and more accurate descisions related to customers, risks, costs and business planning.